

Homework 0

1. Please watch these two YouTube videos before Monday's class. The links are:
<https://www.youtube.com/watch?v=hrTQipWp6co> (37 minutes) and
<https://www.youtube.com/watch?v=RGOj5yH7evk> (68 minutes). The first video focus more on Git and the second talks about both Git and GitHub. There are some redundant topics, but it is good to get a couple perspectives. Feel free to scrub through the videos or watch at faster speeds if you are already quite familiar with Git and GitHub. Also, for the second video, you can skip the part on SSH keys (20:30 through 25:27).

Note that GitHub was updated and the primary branch is now called "main" not "master", so any time you see commands with "master" in them, you'd use "main".

2. Write 2-3 paragraphs that summarizes the first video including important Git commands that were introduced. He then goes through the git installation, but I already did that so I skipped it.

The video starts with explaining what Git is. It gives a version history of the code so we can easily switch between them. We then make a folder with some code in it and change the directory to that folder. Then we initialize git and make a new version.

To make the version we add several files to it and then commit them to finalize the version. We can also retrieve a log of all code too. If we made a mistake we can amend the previous version too. We also use VS code to visualize changes in the staging area and commit them without typing commands. Files can appear in both the staging and changes area because git only tracks changes and not the files themselves. We can also

remove items from the staging area or revert changes either through vs code or via git terminal.

To change to older versions, we can use the checkout command with the commit hash to identify the version to change. Git log will only show old versions unless there is another parameter. When changing old versions, make sure to not make a new branch, but a new version in the same branch. We can also restore individual files back to old versions as well without new branches.

If we want, we can set up aliases to save time. We can also make a .gitignore file to remove some files from version history. We can also remove git from projects too.

3. Write 2-3 paragraphs that summarizes the second video including important Git commands that were introduced (that you did not already mention in for the first video).

A repository is just the folder where your project is. Git hub is where you post git hub repositories online. The clone command can throw stuff on github onto your local machine. Pull can be used for individual changes. Changes can be made online via github too. White means it's the same, red means it is removed, green means it is added. .git files store all changes. Push puts changes on github. Push only works if there is a repository on github to push it to. Pull requests are required if you don't own the project and someone needs to review it. Usually you will add a description to what changes were made and comments on specific lines of code.

Branches may need to be merged so there is continuity between the master and a feature

branch. Specific branches can be specified when pushing to github allowing code to be merged, although merge conflicts can occur if both branches change the same code. Some IDE's like VS Code will make it easy to see the conflict and decide which is the best version to use. If you want to have full control over a project you don't own, you can fork their project and start from there.

4. Submit your summaries to the Homework 0 Assignment on Canvas.